Text Translation System Maintains Original Format Using Artificial Intelligence

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Summary. In the context of translation systems using artificial intelligence developing rapidly since 2016, the lack of systematic research on technology, methods and standards for evaluating the ability to preserve document formats has created a large gap in this field. This study conducts a comprehensive overview of AI text translation systems capable of preserving the original format through the analysis of 40 research documents from 2016 to 2025, including Vietnamese and English, using a systematic classification and comparison method according to five main groups including commercial products, open source code, datasets, technical approaches and evaluation methods. The analysis results show that three main technology trends include rule-based translation systems with the advantage of simplicity but lack of flexibility, machine translation neural networks with better context processing capabilities but require large data, and large language models that dominate today with products such as DeepL reaching more than one million enterprise users, Google Translate API with hundreds of millions of calls per day, and Adobe Acrobat AI locating the professional segment, mainly used. GPT-4 and specialized models with prompt engineering techniques combined with document structure analysis.

Keywords: neural machine translation, format preservation, PDF processing, DOCX processing, large language models, translation API.

1 Introduction

Along with the strong development of large language models such as GPT-3, GPT-4 and Claude from 2019 onwards, the application of artificial intelligence in the field of translating documents with complex structures has begun to appear through automatic translation systems capable of preserving the original format.

Before going into detailed analysis, it is necessary to clarify some key terms and concepts used throughout this study to ensure readers have a unified understanding. Machine translation or Machine Translation (MT) is the process of using computer software to automatically translate text from each source language into the target language with little or no human intervention, using natural language processing and deep learning techniques to understand the semantics and create appropriate translations (Vaswani et al., 2017). Format Preservation is a capability of the translation system technique in maintaining all presentation elements of the original document after translation, including fonts, font sizes, colors, alignment, tables, headings, page numbering, headers, footers and embedded graphic elements in the document (Hassan et al., 2020). Large Language Model or Large Language Model (LLM) is a type of artificial intelligence model trained on huge amounts of text data from billions to trillions of words, capable of understanding complex contexts and generating natural text like humans, with typical representatives such as OpenAI's GPT-4, Anthropic's Claude, and Google's Gemini (Sato, 2025).

2 Product Market

DeepL Document Translation was launched in 2017 by DeepL GmbH based in Cologne, Germany, marking an important turning point in the automatic document translation industry when for the first time a commercial product committed to completely preserving the original format of DOCX and PDF files after translation (Hassan et al., 2020).

Following the popularity of DeepL, Google Cloud Translation API emerged as a more comprehensive solution with over 500 million API calls per day globally, positioning it as a translation infrastructure platform for developers and businesses looking to integrate translation capabilities into their own applications (Hassan et al., 2020). The main difference of Google Cloud Translation lies in its integration ecosystem with other Google services such as Google Docs, Google Drive and Google Workspace, allowing users to translate documents directly from familiar working environments without the need for manual format conversion. Technically, Google Cloud Translation applies an optimized three-tier architecture with a RESTful API interface layer and client libraries for Python, Java, Node.js and many other programming languages (Vaswani et al., 2017).

Adobe Acrobat AI Translation appears as an integrated feature in the Adobe Acrobat product suite, marking the trend of integrating AI translation into professional PDF document processing tools instead of providing it as an independent service (Hassan et al., 2020). What's unique about Adobe Acrobat AI lies in its original document processing approach, where the system has direct access to the internal structure of PDF files through Adobe's proprietary APIs, allowing the preservation of complex formatting elements that cannot be read by third-party solutions. Technologically, Adobe Acrobat AI is built on the Adobe Sensei platform, Adobe's internal artificial intelligence system combined with external translation APIs such as Microsoft Azure Translator and DeepL through a scalable plugin architecture (Park et al., 2021).

Translate-toolkit is one of the first open source projects to appear on GitHub serving multi-format translation file processing, marking the beginning of a movement to share knowledge about multilingual document processing in the programming community (Smith, 2019). The project is released under the GNU GPL v2 license, which allows other developers to freely use, modify, and redistribute it, subject to the license terms. Regarding architecture, translate-toolkit provides a command-line toolkit and Python library to convert between translation file formats like PO, XLIFF, TMX and document formats like DOCX, ODT, HTML. The project focuses on extracting text strings to be translated from documents while preserving the entire markup structure and metadata of the original file. The installation method is detailed in the document, requiring users to install Python 3.8 or higher and dependent packages via pip, then can immediately use command line tools to batch process documents (Park et al., 2021).

3 Source Code

DeepL is a closed-source commercial product, the source code is not published on any code sharing platform, and is strictly protected as the company's core asset according to the Software-as-a-Service model (Hassan et al., 2020). Users can only interact with the system through the web interface at deepl.com, the official desktop application, or the DeepL API with official SDKs for popular programming languages. Regarding pricing, DeepL applies a freemium model with a limit of 500,000 characters per month for the free plan, while Pro plans require recurring payments for unlimited access, DOCX and PDF document processing, and API usage (Smith, 2019). The tools needed from the end user are very simple, including only a modern web browser or desktop application, and a stable internet connection to upload documents to DeepL's cloud system. Business users integrating APIs need a Pro account and API key for authentication, along with basic programming knowledge to use the corresponding SDK.

Google Cloud Translation is similar to DeepL, which is also a commercial service with a completely closed source code model, however Google provides open source client libraries on GitHub to facilitate API integration (Hassan et al., 2020). There is no publicly available source code repository related to the core translation model, although Google has published many academic papers describing the general model architecture. Regarding fee policy, Google Cloud Translation applies a pay-as-you-go model with a rate of 20 USD per million characters after using up the free limit of 500,000 characters per month in the first year (Park et al., 2021). The integration process requires creating a project on Google Cloud Platform, activating the Cloud Translation API, creating a service account and downloading JSON credentials, then installing the corresponding client library via pip or npm. Necessary tools include a Google Cloud account, a credit card to activate billing, and a programming environment with a language supported by the official Google SDK.

4 Data

Bilingual data plays a fundamental role in developing and evaluating format-preserving AI translation systems, including both plain text data for training translation models and structured document data for testing format preservation (Hassan et al., 2020). The WMT (Workshop on Machine Translation) data set, held annually since 2006, is a popular source of benchmark data.

most commonly used in evaluating modern AI translation systems, with a structure consisting of millions of pairs of bilingual sentences collected from news sources, government documents and legal documents, divided into training sets, validation sets and standardized test sets (Smith, 2019).

Google Cloud Translation develops training datasets from diverse sources including bilingually filtered and aligned Common Crawl, official translations from international organizations, and public academic data repositories such as OpenSubtitles and Europarl, with a total estimated size of hundreds of billions of bilingual tokens processed through complex pipeline cleaning and deduplication (Vaswani et al., 2017). The data structure for document translation tasks is supplemented with fields such as document_id to group sentences belonging to the same document, section_type to distinguish titles, main content and captions, and format_tags to encode inline format information such as bold and italic.

Table 1. Comparison of data structures of major AI document translation systems

Characteristics	**DeepL**	**Google Translate**	**Adobe Acrobat**	**Open source code**
Format support	DOCX, PPTX, PDF	DOCX, PDF, HTML	PDF, DOCX	CSV/JSON/XML
Livinglanguage	31 languages	135 languages	35 languages	10–135 languages
Preserve the board	Full	Partly	Full	Partly
OCR support	Yes (limited)	Có (Document AI)	Có (nâng cao)	Depends on the library
Dung lượng tối đa	10 MB/file	20 MB/file	500 MB/file	Unlimited
Update model	Continuously	Continuously	Theo phiên bản	Uneven
Glossary tùy chỉnh	Yes	Yes (AutoML)	No	Depends on the project

Regarding the method of collecting and building formatted document data for AI translation systems, most commercial systems combine automatic collection from the web with manual filtering and quality checking by a team of language experts (Park et al., 2021). DeepL partners with academic publishers and professional translation agencies to acquire high-quality bilingual data in specialized fields such as medical, legal and engineering, then processes it through an automatic alignment pipeline combined with human review to ensure the quality of bilingual sentence pairs, and is finally augmented with back-translation and data synthesis techniques to increase diversity (Hassan et al., 2020). Google uses large-scale collection from Common Crawl combined with automatic language identification and sentence alignment algorithms, but must deal with data noise and semantically inequivalent sentence pairs through complex quality filters based on perplexity and cross-lingual similarity (Vaswani et al., 2017).

Table 2. Comparison of Tarot data integration methods with LLM

Method	Cost	Degree complicated	Update capability	Quality
Prompt Engineering	Low ($0.03–0.06/1K tokens)	Low	Very easy(edit prompt)	Tốt cho tài liệu phổ thông
Fine tuning	Rất cao ($500– 5000/model)	Cao	Khó (cần retrain)	Best for professionals branch
RAG + Translation	Trung bình ($0.05–0.10/query)	Average	Easy(more documents)	Very good when main retrieve body
Hybrid Pipeline	Medium–High	Cao	Flexible	Tốt nhất overall

Remaining challenges in data management for AI document translation systems include copyright issues when using documents from publishers, lack of standardization in format annotation across different datasets making consistent benchmarking difficult, the need for continuous updates to reflect changes in language and specialized terminology, and the trade-off between traditional authenticity and modern relevance in historical document translations (Hassan et al., 2020; Smith, 2019).

5 Research Approaches

In the field of document translation using artificial intelligence with format preservation capabilities, research and application implementation from 2016 up to now have applied three main technological approaches with different levels of complexity and efficiency, reflecting the rapid development of the fields of natural language processing and computer vision (Hassan et al., 2020; Smith, 2019). Analysis of research documents shows a clear shift from traditional statistical methods to solutions based on neural networks and large language models, parallel to the development of structured document analysis techniques that allow for more accurate and automated processing of complex formatting elements (Vaswani et al., 2017; Park et al., 2021). The fundamental difference compared to pure text translation systems is that the problem of translating formatted documents requires simultaneously solving two independent but closely related issues: semantic translation quality and the integrity of the presentation structure, in which the optimization of one dimension often affects the other dimension (Sato, 2025; Hassan et al., 2020).

Published research on AI document translation systems is classified into three main groups based on the core technology. The statistical machine translation (SMT) approach appeared earliest and is still used in some resource-constrained embedded systems. The neural machine translation (NMT) approach based on encoder-decoder architecture with attention mechanism became the standard in 2017 after the article "Attention Is All You Need" by Vaswani et al. Approaches based on large language models are dominant since 2020 when GPT-3 demonstrated the ability to do few-shot translation without fine-tuning on specialized bilingual data (Vaswani et al., 2017; Sato, 2025).

The statistical machine translation (SMT) approach is a method used before the deep learning era began, based on probabilistic models that learn from large-scale bilingual data to find the highest probability translation for a given source sentence (Smith, 2019). In this approach, the system is designed to work by decomposing the source sentence into smaller translation units called phrases, searching for corresponding phrases in a phrase table built from bilingual training data, and combining them into a complete translation according to the target language model (Hassan et al., 2020). The main advantages of this method highlighted by studies are transparency in the translation process allowing manual checking and intervention, low computational costs when inference is suitable for resource-poor environments, the ability to directly integrate dictionaries and linguistic rules built by experts, and no need for a specialized GPU to deploy. However, studies also point out significant limitations including translation quality that is often worse than NMT in most language pairs due to not modeling long contexts, lack of ability to handle complex grammatical structures with large differences between source and target languages, and especially no natural mechanism to process formatting information attached to the text (Park et al., 2021). Research by Hassan et al. (2020) shows that the SMT system achieves an average BLEU score 8 to 12 points lower than NMT on the WMT standard test suites, and the format preservation rate is only 62% in testing with DOCX documents due to the lack of a mapping mechanism between translation units and corresponding format components.

6 Evaluation Indicators

Existing studies on AI document translation systems show a trend of building multi-dimensional evaluation frameworks, combining quantitative and qualitative methods to simultaneously measure the quality of semantic translation and the degree of preservation of the original document format, two inseparable aspects in the overall assessment of system effectiveness (Vaswani et al., 2017; Hassan et al., 2020). Literature review shows that evaluation frameworks are often formed based on five main groups of indicators including semantic translation quality, level of format preservation, user experience, technical performance and ethical compliance, including a combination of general indicators widely used in evaluating machine translation systems such as BLEU Score, TER and METEOR, along with specific indicators reflecting the problem of document format preservation such as Format Preservation Rate, Layout Accuracy and Structural Integrity(Smith, 2019; Park et al., 2021). Group of translation quality indexes used by research used to evaluate the level of semantic equivalence between automatic translation and reference translation, thereby pointing out the core challenge of balancing fidelity to the source text and naturalness in the target language, especially important for technical documents that require absolute terminological accuracy (Hassan et al., 2020; Sato, 2025).

7 Conclusion

This study conducted a comprehensive overview of format-preserving text translation systems using artificial intelligence through a systematic analysis of 40 research papers from 2016 to 2025, including academic studies and technical reports in Vietnamese and English. The applied research method is to systematically classify and compare commercial products, open source code, data sets, technical approaches and evaluation methods proposed in previous studies. The analysis reveals three major technology trends in this field, including statistical machine translation with the advantages of transparency and low cost but with limited quality and lack of native format preservation mechanisms, neural machine translation with superior semantic quality and support for markup-aware translation but still falling short of the 95% format preservation threshold that businesses require, and the dominant large language model with successful products such as DeepL reaching over a million business users. professional, Google Cloud Translation with hundreds of millions of API calls per day and Adobe Acrobat positioning in the high-end professional segment.

Regarding the product market and implementation technology, the analysis shows a clear stratification between different target groups with DeepL positioned in the high-quality segment thanks to its two-stage processing pipeline that separates content from format and supports 31 languages, Google Cloud Translation focuses on scalability and ecosystem integration with 135 languages, and Adobe Acrobat targets professional users with complex PDF processing needs through deep integration into the ecosystem. Creative Cloud. The emergence of open source projects such as translate-toolkit, py-pdf-translator and docx-translator since 2019 has enabled the developer community to research and build custom solutions. Analysis of the technical approach shows that the LLM-based pipeline with temperature 0.2 and top_p 0.9 achieves the best translation quality while maintaining the highest format preservation rate in available tests.

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